Tyler Harnish

Manalapan, New Jersey
Tyler.M.Harnish@gmail.com | +1 (908) 202-4119 | linkedin.com/in/tyler-harnish-th117 | github.com/TylerHarnish5

EDUCATION

Kean University — Union, NJ | May 2026
Bachelor of Science in Computer Science, Magna Cum Laude | GPA 3.76 / 4.0
Relevant Coursework: Python Programming, Deep Learning & Applications, Web Application Development, Database Management Systems, Object-Oriented Analysis & Design, Computer Security, Human Computer Interaction

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, PL/SQL, HTML, CSS
Frameworks & Libraries: React, Next.js, FastAPI, SQLAlchemy, PyTorch, torchvision, scikit-learn, Pandas, Tailwind CSS, Framer Motion
Databases & APIs: PostgreSQL, MySQL, REST APIs, Alembic, Pydantic
Tools & Platforms: Git, GitHub, Docker, Docker Compose, pytest, Ruff, n8n, GitHub Actions, Amazon Web Services (AWS), EC2, RDS

PROJECTS

Residential Clothing Automation Platform
Technologies: Python, FastAPI, AWS, PostgreSQL, SQLAlchemy, Alembic, Docker, pytest, Ruff, n8n
- Built a full-stack Python/FastAPI order-management system backed by PostgreSQL supporting resident budgets, inventory reservation, approval workflows, and partial fulfillment.
- Designed REST APIs and relational models with SQLAlchemy and Alembic; implemented durable idempotency to prevent duplicate writes during request retries.
- Automated CI with GitHub Actions and Ruff, wrote 71+ pytest tests, including integration tests, containerized with Docker Compose, deployed on AWS EC2 with PostgreSQL on Amazon RDS, and integrated n8n order intake.

Portfolio Web Application
Technologies: JavaScript, TypeScript, Next.js, React, PostCSS, Tailwind CSS, Framer Motion, Vercel
- Built a responsive portfolio with Next.js, React, TypeScript, Tailwind CSS, and Framer Motion using a reusable component-based architecture.
- Developed interactive UI components and optimized desktop/mobile layouts; deployed to Vercel through GitHub-based continuous deployment.

TrashNet Waste Image Classification Using Deep Learning
Technologies: Python, PyTorch, torchvision, ResNet18, Pandas, Matplotlib, Hugging Face Datasets
- Fine-tuned a pretrained ResNet18 on 5,054 TrashNet images across six waste classes using transfer learning with PyTorch and torchvision.
- Evaluated dropout, batch normalization, and weight-decay regularization, achieving 95.65% test accuracy through iterative model experimentation.

EXPERIENCE

Technical Assistant **October 2024 – May 2026**
Nancy Thompson Learning Commons, Kean University
- Troubleshot and configured audiovisual and presentation technology in Kean University's V.I.S.E.R. lab, providing hands-on technical support for presentations, university events, and virtual reality sessions.

General Laborer **June 2023 – Present**
Unique Triangle Apparel, Marlboro, NJ
- Supported warehouse inventory organization and on-site shopping operations for assisted-living facilities.